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Inclusive digital competence among Spanish teachers: a comparative study between non-university and university levels

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ABSTRACT

The development of teachers' digital competence is essential to ensuring inclusive education in environments that are increasingly technology-mediated. This study conducts a comparative analysis of the level of digital competence among primary and university teachers in relation to addressing student diversity. A quantitative, comparative, cross-sectional ex post facto design was employed. The sample comprised 777 primary school teachers and 2072 university lecturers, selected through purposive non-probabilistic sampling. Data were collected via online questionnaires using a six-point Likert scale, structured across six dimensions: general, visual, auditory, motor, cognitive and accessibility competence. Data analysis included descriptive statistics, t-tests for independent samples, analysis of variance (ANOVA/ANCOVA), linear regression models and standardised analysis using z-scores. The results show average levels of digital competence in both groups ($M = 3.26$ in primary education; $M = 3.47$ in higher education), with significant differences in favour of university lecturers in the general and cognitive dimensions, and similar scores in accessibility, the weakest dimension in both groups. It is concluded that, despite the greater technological maturity of university lecturers, common challenges related to accessibility and digital inclusion persist. It is proposed that teacher training in inclusive digital competence be strengthened through contextualised pedagogical strategies.

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Digital competence in teaching; educational inclusion; higher education; attention to diversity; educational technology

Introduction

The digital transformation of the education system has made digital competence in teaching an essential element in ensuring high-quality, inclusive and equitable education. Over the last two decades, technological advances have changed not only teaching methodologies, but also the concept of accessibility and student participation in learning environments. This evolution requires teachers to develop comprehensive digital competence, combining technical mastery of technologies with the ability to use them in a pedagogically inclusive manner (Redecker, 2017; UNESCO, 2018).

Far from being merely a set of technical skills, teachers' digital competence encompasses ethical, social and pedagogical dimensions, aimed at creating accessible and inclusive learning environments, understood as educational contexts designed in accordance with the principles of Universal Design for Learning (UDL), which promote accessibility, flexibility and the participation of all learners by providing multiple forms of representation, action, expression and engagement (CAST, 2024; Ferrari, 2013; Ilomäki et al., 2016). From this perspective, the concept of inclusive digital competence emerges, understood as the ability of teachers to integrate digital technologies in order to cater for the diversity of their students, promoting the full participation of students with disabilities or specific educational support needs (Cabero-Almenara et al., 2021; Castañeda & Selwyn, 2018). This concept arises in response to two persistent challenges in contemporary education systems: the digital divide, which limits equitable access to technological resources, and the accessibility divide, which excludes certain groups from the effective use of ICT.

Recent literature has emphasised that the pedagogical use of technology should be geared towards equity and the elimination of barriers (CAST, 2018; Tang et al., 2024). From this perspective, digital competence in teaching is understood as a multidimensional process that involves knowing, knowing how to do

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and knowing how to be in diverse technological contexts (Redecker, 2017). However, several studies have pointed out that the most widely used reference frameworks, such as DigCompEdu (European Commission, 2022), although useful for describing teachers' digital performance, do not explicitly incorporate the inclusive dimension (Bong & Chen, 2024; Fernández-Batanero et al., 2022b). For this reason, complementary approaches have been developed that integrate accessibility and UDL as central axes of digital competence (Cabero-Almenara et al., 2021; Fernández-Batanero et al., 2022a).

Within the European context, the development of teachers' digital competence and inclusive education is underpinned by a set of regulatory and strategic frameworks that guide current education policies. Among these, the European Framework for the Digital Competence of Educators (DigCompEdu) stands out, establishing a comprehensive model of teachers' digital competence, including dimensions related to accessibility, inclusion and the personalisation of learning. Similarly, the Digital Education Action Plan (2021–2027) promotes the use of digital technologies to ensure more inclusive and resilient education systems, emphasising the importance of bridging the digital divide.

Several studies have shown that Spanish primary school teachers have limited knowledge of ICT applied to students with disabilities, and that most have not received specific training in this area. However, a similar phenomenon has also been observed among university teachers: although their overall level of digital competence is high, their training in the use of digital resources for inclusion is insufficient (Barajas Motta et al., 2025; Fernández-Batanero et al., 2019; Fernández-Batanero et al., 2022b). This reflects a consistent pattern: the existence of a gap between technological availability and actual inclusive competence. In other words, teachers are proficient in the technical use of digital tools, but not necessarily in their potential to reduce inequalities or address functional diversity. This shortcoming highlights the need to reconceptualise teacher digital literacy as a process that integrates the ethical and social dimensions of technology (Heitink et al., 2016; Rossi Cordero & Barajas Frutos, 2018).

This finding can be interpreted in the light of the TPACK (Technological Pedagogical Content Knowledge) model, which posits that the effective integration of technology into teaching does not depend solely on technical proficiency, but on the interaction between technological, pedagogical and content knowledge (Kim, 2024; Li et al., 2025). From this perspective, the findings suggest that teachers may possess an adequate level of technological knowledge but face difficulties in integrating it with pedagogical approaches geared towards inclusion. In particular, limited competence in accessibility and attention to diversity would indicate a weakness in the pedagogical component of the model, specifically in the ability to apply technology based on inclusive principles, such as the DUA. Thus, the gap identified between general digital competence and inclusive digital competence could be explained as a lack of effective integration between the different domains of teaching knowledge, rather than as an exclusively technical shortcoming.

Inclusive digital competence is configured differently according to educational level, training opportunities and available institutional resources (Cabero-Almenara et al., 2021). In primary education, digitisation policies have promoted the incorporation of ICT into the classroom as a tool for diversifying teaching strategies and adapting them to the heterogeneity of the student body. However, teachers at this level tend to have less technical support and less autonomy to design inclusive resources or implement assistive technologies (Montenegro-Rueda et al., 2023). In contrast, in university education, the technological infrastructure is more robust and digital integration more advanced, but with an often instrumental and administrative approach, focused on the management of virtual learning (Martínez Márquez et al., 2025). Despite this technical advantage, studies such as that by Fernández-Batanero et al. (2022a) show that the inclusive digital competence of university teachers does not significantly exceed that of teachers at previous levels, indicating a cross-cutting gap in teacher training.

Despite advances in infrastructure and educational digitisation, structural gaps persist that affect teachers' ability to integrate ICT from an inclusive perspective, such as a lack of training in digital accessibility (Fernández-Batanero et al., 2022a). Overcoming these gaps involves rethinking the relationship between technology and inclusion from an ethical and transformative perspective. It is not just a matter of incorporating digital tools, but of ensuring that these tools promote the full and equitable participation of all students (UNESCO, 2023).

Furthermore, in recent years, the rise of generative artificial intelligence (generative AI) has brought about a significant shift in digital educational environments, particularly in terms of accessibility and inclusion. These technologies enable the automated creation of adapted teaching materials, which can promote the

participation of students with diverse needs (Fernández Cerero et al., 2025: Secanell et al., 2025; Wang et al., 2025). Furthermore, recent studies highlight that generative AI can facilitate the design of more inclusive learning experiences by supporting the cognitive, linguistic and sensory adaptation of content (Lopez-Gazpio, 2025).

However, the literature also warns of significant risks, such as the reproduction of algorithmic biases, a lack of transparency in content generation processes, or the creation of materials that do not meet universal accessibility criteria (Caboni & Pizzichini, 2025). In this context, the concept of "inclusive AI" emerges, emphasising the need to design, implement and use artificial intelligence systems based on principles of equity, accessibility and educational justice, incorporating a critical perspective into teachers' digital competence.

This study provides a comparative and standardised view of the perceived inclusive digital competence of Spanish teachers at two strategic educational levels: primary and university. Unlike previous studies that focused on a single level, this analysis identifies cross-cutting patterns and common gaps, offering useful empirical evidence for the design of teacher training policies and institutional strategies aimed at digital accessibility and educational equity.

Unlike previous research, which has largely focused on analysing teachers' digital competence at a single educational level or without explicitly incorporating the dimension of inclusivity, this study makes three specific contributions. Firstly, it offers a direct comparative analysis between primary and higher education teachers using equivalent and metrice-invariant instruments. Secondly, it systematically incorporates the dimension of inclusive digital competence, operationalised across six specific areas (general, visual, auditory, motor, cognitive and accessibility), enabling the precise identification of areas of greatest weakness. Thirdly, it combines inferential analyses and standardisation procedures to ensure the robustness and comparability of results across different educational contexts. These contributions advance our understanding of the gap between general digital competence and perceived inclusive digital competence, an aspect scarcely addressed in the previous literature.

Our study therefore aims to contribute to a deeper understanding of the link between digitalisation and inclusion, emphasising that true technological innovation is only possible when it is built on the principles of justice, participation and universal accessibility. In this regard, we consider this work to be of particular importance because it takes an empirical and comparative approach to the perceived inclusive digital competence of teachers, a key area for ensuring equitable education in increasingly widespread technological environments. Unlike most previous research, which has focused on a single level of education, this study simultaneously analyses the contexts of primary and university education, allowing us to identify cross-cutting patterns and common gaps in teacher preparation for addressing diversity.

Purpose and research questions

The overall objective of this study is to compare the level of perceived digital competence in teaching for diversity among primary and university teachers, analysing overall differences and differences in specific dimensions.

The research questions that guided the study were as follows:

(1) What is the overall level of perceived digital competence in teaching among primary and university teachers in terms of diversity among students?
(2) Are there significant differences between the two educational levels in the six dimensions of perceived inclusive digital competence (general, visual, auditory, motor, cognitive and accessibility)?
(3) What personal and professional variables (age, years of experience, school tenure, previous training in inclusive ICT) influence the level of digital competence in teaching in each group?
(4) To what extent do the results remain consistent after data standardisation, and what implications does this have for comparability between different educational contexts?
(5) What common gaps and challenges are identified in the integration of technology with an inclusive approach in primary and university education?

Method

Research design

This study adopts a quantitative, comparative and cross-sectional design, aimed at the descriptive and inferential analysis of levels of perceived digital competence among teachers in two different educational contexts: primary education and university education.

An ex post facto causal-comparative design was chosen, given that the independent variables (educational stage, age, experience, school tenure) were not manipulated by the researcher but observed as they occur in reality (Creswell & Creswell, 2017). The methodological approach combines the analysis of means and variances for comparison between groups with regression models to explore the influence of sociodemographic variables. It also includes a robustness analysis through data standardisation (z-scores) to verify the consistency of the differences observed.

Participants

The total sample consisted of 2849 teachers from two educational levels:

-- Primary school teachers (n = 777): Professionals working in primary schools in different institutional contexts (public and private) in Spain.
-- University teachers (n = 2072): Teachers from different disciplinary areas in higher education institutions at Spanish universities.

The sample was selected using non-probabilistic, purposive sampling, specifically through criteria-based sampling. This approach involved selecting participants who met specific criteria relevant to the study's objectives, such as currently working as teachers, having at least one year's professional experience, and having participated in teaching processes facilitated by digital technologies during the previous academic year. Participants were recruited through professional networks and voluntary participation. Therefore, teachers with greater interest in digital technologies and inclusion may have been more likely to participate. This self-selection bias could have resulted in an overestimation of perceived competence levels and limits the representativeness of the sample.

This type of sampling is appropriate for comparative educational studies, where the aim is to analyse specific profiles of participants with relevant experience in the phenomenon under study, rather than to obtain a statistical representation of the general population.

Part of the sample used had already been analysed in a previous study; its inclusion in this study is based on the fact that it allows for a comparison of the perceived inclusive digital competence of university lecturers and that of primary school teachers, providing a comparative perspective across different stages of education that facilitates a deeper understanding of training needs, strengths and challenges associated with the use of digital resources to support diversity (Fernández-Cerero & Román-Graván, 2023)

Participants were recruited through purposive sampling based on professional and academic networks. In total, invitations were sent to approximately 1900 primary school teachers and 4700 university lecturers via institutional mailing lists, email and professional associations. Ultimately, 777 valid responses were received from the primary school group and 2072 from the university group, representing a response rate of approximately 40.9% and 44.1%, respectively.

In the primary school group, the average age was 42.3 years (SD = 8.9) and 68.5% were women. In the university group, the average age was 45.6 years (SD = 9.7), with 59.4% women. The average teaching experience was 14.8 years (SD = 9.2) in primary education and 17.3 years (SD = 10.4) in university education. Table 1 presents an overview of the sample.

Data collection instruments

Two parallel questionnaires were used for data collection, designed with the same conceptual and metric structure, but adapted to the context and language of each educational level. Both instruments were previously validated using confirmatory factor analysis and showed high internal reliability in all dimensions.

Table 1. Características de la muestra de docentes de primaria y universidad.

Variable	Primary (n = 777)	University (n = 2072)
Mean Age (SD)	42.3 (8.9)	45.6 (9.7)
Women (%)	68.5	59.4
Teaching experience (Years, SD)	14.8 (9.2)	17.3 (10.4)
Public sector employment (%)	71.2	65.8
Previous training in inclusive ICT (%)	46.9	38.1

The questionnaires assess Digital Teaching Competence for Attention to Diversity (CDDAD) across six dimensions:

- General competence: use of digital technologies to plan, teach and assess in an inclusive manner.
- Visual: knowledge and application of digital resources for students with visual impairments (screen readers, contrast, accessible fonts).
- Auditory: adaptation of materials through subtitling, transcription and auditory support tools.
- Motor: design of accessible digital environments for students with reduced mobility.
- Cognitive: digital strategies and resources to support the comprehension, memory and attention of students with cognitive difficulties.
- Accessibility: mastery of universal accessibility principles and UDL in digital environments.

Both instruments use a 6-point Likert scale, ranging from 1 (very low mastery) to 6 (very high mastery).

- Primary School Questionnaire: 71 items. Overall reliability α = .961; reliability by dimension between α = .930 and .950.
- University questionnaire: 56 items. Overall reliability α = .939; reliability per dimension between α = .917 and .947.

The composite reliability values (Ω) ranged from .925 to .949, confirming the internal consistency of both versions. Table 2 summarises the dimensions, number of items and reliability coefficients.

The information collection instruments are available at the following links:

- First instrument (Primary): https://docs.google.com/forms/d/e/1FAIpQLSfux6m1cU6Nf-69eiiMS28LjcSom38yqe2OmS-Jy4mXAgJVnA/viewform
- Second instrument (University): https://docs.google.com/forms/d/e/1FAIpQLSeTBRvQk1fHLLJyJOwPZ4TVsoLoXvH_T6UjVa0zjq5hi6FjTw/viewform

Validity and reliability of the instruments

The content validity of the instruments was ensured through review by expert judges specialising in inclusive education and educational technology (n = 8), who assessed the relevance, clarity and consistency of the items in relation to the theoretical dimensions of the construct. Construct validity was examined using confirmatory factor analysis (CFA), yielding adequate fit indices (CFI = .962; TLI = .955; RMSEA = .041), which support the proposed factor structure.

Table 2. Dimensions, number of items and reliability of the instruments.

Dimension	No. of items primary school	No. of item university	Cronbach's α	McDonald's Ω
General	15	12	.932	.925
Visual	12	9	.937	.928
Auditory	10	8	.945	.931
Motor	9	8	.930	.927
Cognitive	13	10	.948	.940
Accessibility	12	9	.941	.937
Total	71	56	.961	.939

Furthermore, a multi-group factorial invariance analysis was conducted to verify metric equivalence between the versions of the questionnaires administered in primary and higher education. The results confirmed scalar invariance, allowing for a valid comparison of scores between the two groups. In terms of reliability, both instruments demonstrated high levels of internal consistency. In the primary education questionnaire, the overall reliability was $\alpha = .961$, with values per dimension ranging from $\alpha = .930$ to .950. In the university education questionnaire, the overall reliability was $\alpha = .939$, with values ranging from $\alpha = .917$ to .947. The composite reliability coefficients (Ω) ranged from .925 to .949, confirming the robustness of the measures used.

Procedure

The fieldwork was carried out over a period of six months. The questionnaire was distributed in digital format via an online survey platform. The invitation to participate was disseminated through institutional networks and professional associations. The invitation to participate was circulated via institutional email, professional teaching networks and educational associations, including direct links to the online questionnaire. Regular reminders were sent out to encourage participation. The dissemination strategy combined direct contact with educational institutions and open distribution via professional networks, which enabled a broad and diverse sample to be reached.

Before accessing the questionnaire, participants read an information consent form, which guaranteed data confidentiality, anonymity, and voluntary participation. It should also be noted that the average response time was 18min. No significant data loss was recorded. Incomplete responses (2.3% of the total) were eliminated prior to analysis.

Data analysis

Statistical analysis was performed using SPSS v.29 and AMOS v.28. The following phases were developed:

1. Descriptive analysis: calculation of means, standard deviations, and frequency distribution for each dimension of CDDAD.
2. Comparison between groups: Student's *t*-test for independent samples. Factorial ANOVA and ANCOVA, controlling for age, experience and tenure.
3. Effect size: calculation of Cohen's d and Hedges' g with 95% confidence intervals.
4. Predictive models: multiple linear regression with overall digital competence as the dependent variable and sociodemographic predictors.
5. Standardised analysis: scores were transformed into *z*-scores ($M = 0$, SD = 1) within each group, repeating the comparisons to confirm the robustness of the results.
6. Verification of assumptions: normality (Kolmogorov–Smirnov), homogeneity of variances (Levene) and multicollinearity (VIF < 5).

Ethical considerations

This study was conducted in accordance with the ethical principles set out in the Declaration of Helsinki and in line with standard guidelines for educational research involving human subjects. Approval was obtained from the Ethics Committee of the XXXXX. Participation in the study was entirely voluntary. Before accessing the questionnaire, all participants received an information sheet explaining the objectives of the research, the data collection procedure and the intended use of the information. Informed consent was obtained electronically before participants began to complete the questionnaire.

Participants were assured of their right to withdraw from the study at any time without this entailing any negative consequences. No incentives were offered for participation. To ensure anonymity, no personally identifiable information was collected. All responses were recorded anonymously and the data were stored in password-protected files, accessible only to the research team. Furthermore, the confidentiality of the information collected was guaranteed. The data were used exclusively for academic and research purposes, and the results are presented in aggregate form to prevent the individual identification of participants.

Results

The results of the study provide a detailed description of the competency profile of the participating teachers and the differences observed between primary and university education contexts in relation to teachers' digital competence for addressing diversity.

Overall level of teachers' digital competence

The descriptive analysis indicates that both primary and university teachers have intermediate levels of perceived inclusive digital competence. On the 1–6 point Likert scale, the overall average was $M = 3.26$ (SD = 0.87) for primary school teachers and $M = 3.47$ (SD = 0.81) for university teachers.

The difference between the two groups is statistically significant (t (2847) = −7.28, $p < .001$), with a moderate effect size (Hedges' $g = 0.33$), suggesting a competency advantage for university teachers in their self-perception of inclusive technological mastery Table 3.

The difference between groups is consistent across all dimensions except Accessibility, where both groups have low values ($M < 3.0$) and no significant differences. This coincidence confirms that digital accessibility is the weakest and most problematic dimension for both educational levels.

Figure 1 illustrates the average profile of perceived teaching digital competence by dimension and educational stage.

Differences by sociodemographic variables

To examine the factors associated with perceived global digital competence, analyses of covariance (ANCOVA) were performed, including age, years of experience, school ownership, and gender as covariates.

Table 3. Means, standard deviations and contrasts by dimension of digital teaching competence.

Dimensión	Primary M (SD)	University M (SD)	*t*	*p*	Hedges' *g*
General	3.48 (0.92)	3.74 (0.85)	−6.32	<.001	0.30
Visual	3.12 (0.96)	3.38 (0.91)	−5.89	<.001	0.28
Auditory	3.31 (0.89)	3.44 (0.84)	−3.17	.002	0.15
Motor	3.21 (0.95)	3.39 (0.90)	−4.21	<.001	0.21
Cognitive	3.42 (0.90)	3.68 (0.84)	−6.11	<.001	0.29
Accessibility	2.87 (0.97)	2.92 (0.94)	−1.14	.253	0.04
Total	**3.26 (0.87)**	**3.47 (0.81)**	−7.28	**<.001**	0.33

Note: Likert scale 1–6. Higher values indicate greater self-perception of inclusive digital competence.

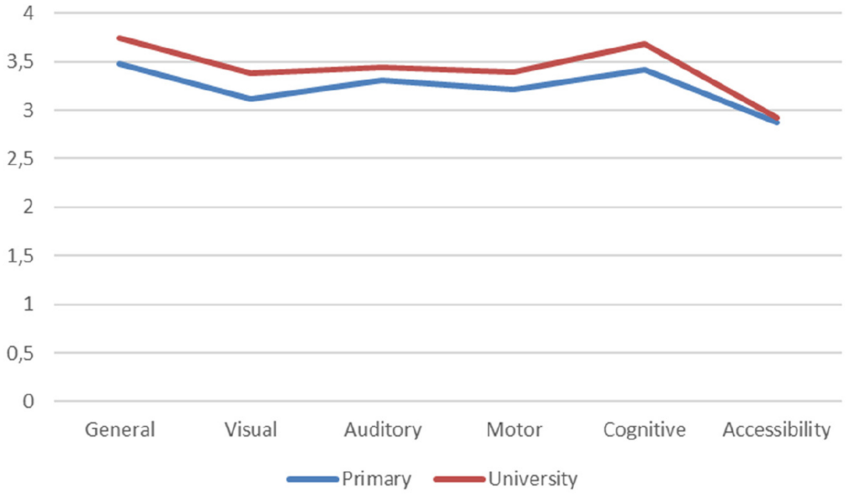


Figure 1. Average profile of digital teaching competence by dimension and educational stage.

Table 4. ANCOVA for perceived global digital competence by educational stage and covariates.

Source	*F*	*p*	Partial η^2
Educational stage (Primary/University)	26.74	<.001	.041
Age	2.14	.143	.001
Teaching experience	8.22	.004	.009
School tenure	5.19	.023	.006
Gender	1.67	.196	.001

Note. Covariates included simultaneously. Partial η^2 indicates the proportion of variance explained by each factor.

The results reveal that the educational stage has a significant effect on perceived global digital competence (F (1,2846) = 26.74; $p < .001$; partial $\eta^2 = .041$).

Age shows no significant influence ($p = .143$), while teaching experience correlates positively, albeit weakly ($r = .16$; $p < .01$). School ownership has a slight effect ($p = .023$), with slightly higher scores in public institutions. Gender is not significant after controlling for the other covariates ($p = .196$).

These results indicate that, once personal and work characteristics are controlled for, the educational context (primary or university) remains the main differentiating factor Table 4.

Distribution of proficiency levels

In order to interpret the results pedagogically, the scores were grouped into three levels: Low (1.0–2.9), Medium (3.0–4.4) and High (4.5–6.0).

In primary education, 58.7% of teachers are at the medium level, 26.9% at the low level and 14.4% at the high level. In university, 62.5% are at the medium level, 21.1% at the low level and 16.4% at the high level Figure 2.

Analysis of differences by dimension

Univariate analyses by dimension (ANOVA) allow us to refine the differences observed.

In general competence, university lecturers' scores are significantly higher (F (1,2847) = 39.95; $p < .001$), indicating better management of resources and technological strategies for planning and evaluating teaching.

In the cognitive dimension, the differences also favour the university (F (1,2847) = 34.81; $p < .001$), confirming greater mastery in the use of technologies that support attention, memory, and comprehension.

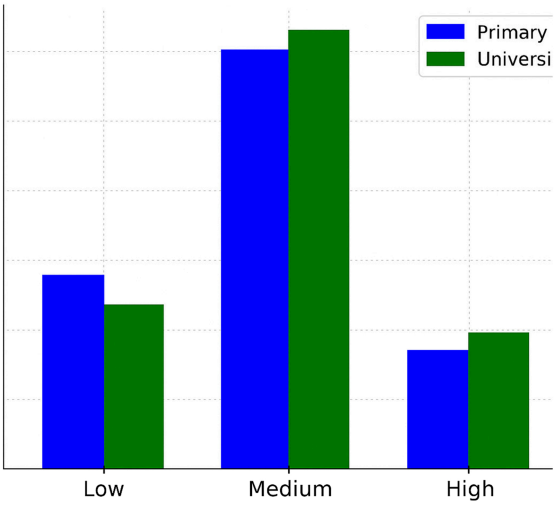


Figure 2. Distribution of levels of perceived digital competence among teachers by educational stage.

In the visual (F (1,2847) = 29.15; $p < .001$) and motor (F (1,2847) = 17.72; $p < .001$) dimensions, the differences are smaller but consistent, reflecting a slight advantage for university teachers in the application of sensory and motor adaptations.

In the auditory dimension, the contrast is marginally significant (F (1,2847) = 7.96; $p = .005$), while accessibility shows no differences (F (1,2847) = 1.14; $p = .253$).

The pattern of results suggests greater technical competence among university teachers and less practical application of accessibility in both groups.

Predictive models

To explore the factors that explain perceived global digital competence, two multiple linear regression models were estimated, one for each educational level.

In primary education, the overall model was significant (F (4,772) = 18.54; $p < .001$) and explained 11.2% of the variance ($R^2 = .112$). The significant predictors were teaching experience ($\beta = .21$; $p < .001$) and previous training in inclusive ICT ($\beta = .29$; $p < .001$). Neither age nor tenure showed relevant effects.

In university education, the model was also significant (F (42067) = 24.91; $p < .001$), with an explained variance of 12.7% ($R^2 = .127$). In this case, the significant predictors were training in inclusive ICT ($\beta = .34$; $p < .001$) and institutional tenure ($\beta = .18$; $p = .017$).

Experience and age did not reach statistical significance Table 5.

At both levels, specific training in inclusive technologies is the most relevant predictor, suggesting that professional training aimed at digital inclusion has a stronger impact than personal characteristics or accumulated experience.

Standardised analysis and robustness

Given that the instruments used vary in length (71 items in primary school and 56 in university), a robustness analysis was performed using intra-group standardisation of scores (z-scores) to verify the stability of the differences found.

After the transformation, the results remained virtually identical:

- Total z-mean = −0.23 (primary school) and +0.18 (university); t (2847) = −7.12; $p < .001$.
- Intraclass correlation (ICC) between original and standardised scores = .987.

These values show almost perfect consistency between the raw and standardised results, confirming that the differences detected are real and not attributable to the structure or length of the instruments. Therefore, the competency gap observed reflects actual differences in teachers' self-perception and not measurement artefacts.

Overall, the results show that university teachers have a stronger competency profile in terms of technical and cognitive management of digital technologies, but share with primary school teachers a cross-cutting weakness in accessibility and inclusive design, which constitutes a structural challenge for teacher training at both levels, as can be seen in Figure 3.

Table 5. Multiple linear regression models by educational level.

Predictor variable	Primary school β (p)	University β (p)
Age	.04 (.214)	.03 (.326)
Teaching experience	.21 (<.001)	.09 (.073)
School tenure	.06 (.187)	.18 (.017)
Training in inclusive ICT	.29 (<.001)	.34 (<.001)
Adjusted R^2	**.112**	**.127**

Note. Dependent variable: Total digital competence. All β values standardised.

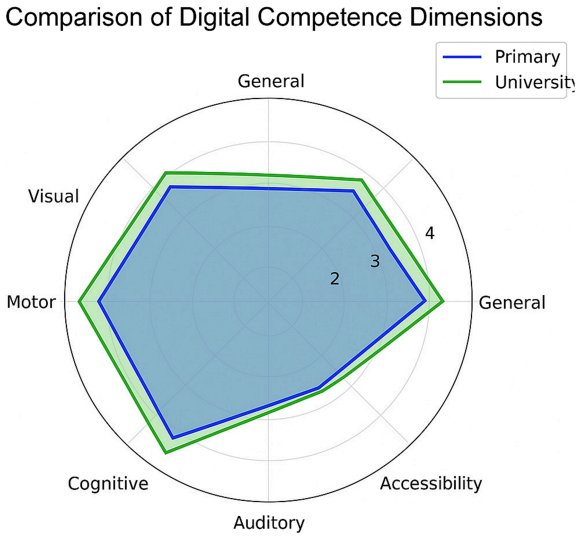


Figure 3. Comparative global representation of perceived digital teaching competence.

Discussion

The results provide a broad overview of the current state of perceived inclusive digital competence at two key educational levels and provide relevant evidence to guide teacher training in digital accessibility and equity.

Overall level of inclusive digital competence among teachers

In relation to the first research question, the results indicate that both primary school and university teachers have average levels of perceived inclusive digital competence, with overall scores of 3.26 and 3.47 respectively on a scale of 1–6. This trend shows a moderate self-perception of technological mastery geared towards inclusion, coinciding with previous studies that indicate that the integration of ICT into teaching processes does not necessarily translate into inclusive pedagogical practices (Cabero-Almenara et al., 2021; Fernández-Batanero et al., 2022c).

Most teachers are concentrated at intermediate levels of competence, while less than 20% reach high levels, confirming that the incorporation of technology in teaching continues to be predominantly instrumental. In this sense, technology is used more as a teaching support tool than as a means of ensuring accessibility or addressing the functional diversity of students (Castañeda & Selwyn, 2018).

These findings underscore the need to conceive of perceived inclusive digital competence not only as a set of technical skills, but also as an ethical and pedagogical competence, oriented towards equity, participation and accessibility (UNESCO, 2018). Therefore, teacher training is required that explicitly integrates the UDL approach and digital accessibility as structural dimensions of technological literacy.

Differences between educational stages and competency profiles

With regard to the second research question, the results show statistically significant differences between the two educational levels, with consistently higher scores among university teachers, especially in the general, cognitive and visual dimensions. These differences, although moderate (Hedges' *g* between .15 and .33), reflect greater technological exposure in the university environment, where teaching relies on virtual learning environments, advanced digital resources and institutional management systems.

These results can be interpreted within the framework of the DUA, particularly in relation to the principle of representation, which promotes the presentation of information in multiple formats to facilitate learners' understanding. In this regard, the cognitive dimension assessed in the study aligns directly with these guidelines, which emphasise the need to offer alternatives in the way content is presented to cater for students' cognitive diversity (CAST, 2024; Pérez-Pérez, 2024).

From this perspective, the differences observed in this dimension could reflect a greater capacity among university lecturers to use digital tools that diversify the representation of information, such as visual resources, graphic organisers or interactive materials. However, the fact that the scores are not high in absolute terms suggests that the systematic application of UDA principles remains limited, reinforcing the need to integrate this pedagogical approach into teacher training in perceived inclusive digital competence (Moriña et al., 2025).

Although the observed differences were statistically significant, their magnitude was generally small to moderate (Hedges' g = .15–.33). Therefore, these findings should be interpreted with caution. Given the large sample size, relatively small differences may reach statistical significance without necessarily implying substantial practical differences. Consequently, the results suggest modest variations in perceived inclusive digital competence rather than large disparities between educational stages.

However, primary school teachers present a more balanced profile across dimensions, which can be explained by their direct contact with a diverse student body and the implementation of digitisation policies in schools (Montenegro-Rueda et al., 2023). This balance suggests a more pedagogical understanding of the use of ICT, albeit with limitations in advanced technical management.

The results are consistent with previous research indicating that competency differences between educational levels respond more to work contexts and institutional culture than to academic level per se (Román-Graván et al., 2024). Thus, the advantages observed in university teachers do not necessarily imply a higher degree of digital inclusion, but rather greater familiarity with general technological tools. Perceived inclusive digital competence, therefore, depends more on cross-cutting factors, such as training and institutional support, than on educational level or accumulated experience.

Interpretation based on the TPACK model

From a theoretical perspective, these results can be interpreted in the light of the TPACK model, which posits that the effective integration of technology into teaching depends on a balanced interaction between technological, pedagogical and content knowledge. In this regard, the data suggest that teachers, particularly at university level, demonstrate a stronger grasp of the technological component, but limited integration with pedagogical knowledge geared towards inclusion (Gatete, 2026).

This interpretation is consistent with recent research which has indicated that mastery of digital tools does not guarantee their effective application in inclusive educational contexts unless accompanied by a pedagogical approach that incorporates principles of accessibility and attention to diversity (Kim, 2024; Li et al., 2025). Thus, the lower scores in dimensions such as accessibility or cognitive adaptation can be understood as a weakness at the intersection between the pedagogical and technological components of the TPACK model.

In this regard, the results of the present study reinforce the need to move towards teacher training models that do not focus exclusively on the development of technical skills, but rather promote a coherent integration of technology, pedagogy and inclusion (Gatete, 2026; Li et al., 2025). From this perspective, perceived inclusive digital competence can be interpreted as an extension of the TPACK model in educational contexts geared towards equity.

Personal and professional factors associated with perceived inclusive digital competence

In relation to the third research question, multiple linear regression models confirm that prior training in inclusive ICT is the strongest predictor of perceived digital competence among teachers in both primary (β = .29; $p < .001$) and university (β = .34; $p < .001$) settings. This finding highlights the importance of specific training in inclusion-oriented technologies, above personal factors such as age or gender, which did not show significant effects.

Teaching experience also has a positive influence, although with less weight and more evident in primary school teachers. In contrast, tenure at the institution has a slight but significant impact in the university context, which could be related to greater technological resources or differentiated institutional policies in public universities.

These results reinforce the idea that perceived inclusive digital competence is built primarily on training opportunities and the professional environment, rather than demographic variables. Consequently, strengthening initial and continuing training programmes focused on accessibility, UDL and assistive technologies is a key factor in the development of digitally inclusive teaching practices (Fernández-Batanero et al., 2022c).

Nevertheless, the explanatory power of the regression models was relatively limited (R^2 = .112 in primary education and R^2 = .127 in university education). Although these values are acceptable within educational research, they indicate that a substantial proportion of the variance remains unexplained. This finding suggests that additional personal, institutional, contextual and technological factors not included in the present study may also play a relevant role in shaping teachers' perceived inclusive digital competence.

Robustness and comparability of results after standardisation

The fourth research question focused on examining the consistency of differences after data standardisation. The z-score analysis confirmed that the patterns of results remain stable, with an intraclass correlation of .987 between the original and standardised scores. This shows that the differences observed between groups are robust and not attributable to the structure or differential length of the instruments applied.

The standardised profile reveals a more homogeneous distribution among primary school teachers and greater heterogeneity among university teachers, with particularly high scores in the general and cognitive dimensions and low scores in accessibility and visual dimensions. This configuration is consistent with studies that describe the digital competence of university teachers as more technical than pedagogical (Moriña & Carballo, 2018). In contrast, the balance among primary school teachers could be due to a more contextualised integration of technology in direct teaching and in the attention given to students with diverse needs.

Overall, the robustness of the results supports the comparative validity between educational contexts and reinforces the interpretation of the differences as genuine and not artefactual.

Common gaps and challenges in inclusive technology integration

The fifth research question aimed to identify shared gaps and challenges in the integration of technologies with an inclusive approach. The results are conclusive in pointing out that digital accessibility is the weakest dimension at both educational levels, with averages below 3.0 and no significant differences between groups.

However, this study does not allow us to determine precisely whether this weakness in the area of accessibility stems from a lack of knowledge among teaching staff regarding technical accessibility standards or from limitations in the use of digital authoring tools that facilitate the creation of accessible materials. This distinction is particularly relevant, as recent research indicates that difficulties with digital accessibility often stem both from a lack of awareness of regulatory principles and from a lack of practical skills to apply these standards using specific tools (Cabero-Almenara et al., 2021; Fernández-Batanero et al., 2022a).

In this vein, the literature highlights that accessibility in digital educational environments does not depend exclusively on the individual competence of teachers, but also on structural factors, such as the training received, the availability of accessible technological resources, and institutional support for their implementation (García-Peñalvo et al., 2020; Montenegro-Rueda et al., 2023). Therefore, the weakness identified in this study could be interpreted as the result of a combination of training gaps and technological limitations, which reduces teachers' ability to translate accessibility principles into educational practice.

This shortcoming reflects a structural weakness in teacher training and practice, coinciding with research that identifies accessibility as the least developed component of digital competence at the European level (García-Peñalvo et al., 2020; Pérez-Pérez, 2024). The limitations detected in the visual and auditory dimensions reinforce this trend, highlighting teachers' lack of familiarity with resources such as screen readers, subtitling, colour contrast, and accessible formats.These gaps suggest a disconnect between the general use of technology and its inclusive application, creating an "internal gap" within teachers' own digital competence (Rossi Cordero & Barajas Frutos, 2018).

This gap cannot be attributed solely to individual motivation, but to structural factors such as the absence

of formal training in digital accessibility, the lack of a cross-cutting inclusive approach in digital competence frameworks, and the lack of specialised technical support in educational centres. Therefore, overcoming these limitations requires a sustained institutional and political strategy aimed at incorporating digital accessibility as a mandatory requirement in teacher training, skills assessment, and the design of digital educational resources.

In this context, the recent introduction of generative artificial intelligence tools represents a key development in the evolution of inclusive digital literacy. These technologies offer significant opportunities to improve accessibility, such as the automatic generation of adapted content, the creation of multimodal materials, or the personalisation of learning according to students' needs (Marengo et al., 2024; Zawacki-Richter et al., 2024). However, their integration into teaching practice requires an additional level of critical and pedagogical competence, as inappropriate use can reproduce biases, generate inaccessible materials or widen the digital divide. Therefore, teachers' inclusive digital competence must be expanded to include the knowledge and responsible use of AI tools (Pagliara et al., 2024), in line with the emerging principles of so-called "inclusive AI".

Conclusions

The study enabled a comparison of the level of perceived digital competence in teaching for diversity among primary and university teachers, identifying common patterns and specific differences. The results suggest that perceived inclusive digital competence is at medium or medium-low levels in both educational contexts, with no evidence of advanced proficiency or cumulative progression throughout one's career. This suggests that the development of this competence does not depend on educational level or experience, but on structural factors linked to training and institutional culture.

The differences between stages are moderate but consistent, with slightly higher scores among university teachers, especially in the general and cognitive dimensions. In contrast, digital accessibility is confirmed as the weakest and most homogeneous dimension in both groups, reflecting a persistent gap between technological integration and its inclusive application. This pattern shows that, although digitisation is advancing, the incorporation of accessibility and universal design into teaching practice remains limited.

Analysis of the associated factors indicates that specific training in inclusive technologies is the main predictor of perceived digital competence among teachers, surpassing variables such as age, gender and professional experience. This finding reinforces the importance of ongoing, context-specific training as a priority means of strengthening perceived inclusive digital competence. Overall, the results reveal a structural gap between teachers' technological competence and inclusive competence, which is shared by both levels of education. Bridging this gap requires integrated training policies, institutional leadership and an ethical conception of technology as an instrument of educational equity and justice. Only then will it be possible to move towards a truly inclusive digital education model, capable of ensuring the full participation and learning of all students at all levels of the education system.

Implications for teacher training and education policy

The results of this study underscore the need to integrate digital accessibility and UDL as mandatory components of initial and continuing teacher training. Teacher training should not be limited to the technical use of ICT but should be geared towards the inclusive pedagogical application of technologies.

It is also recommended that educational institutions develop coherent training and organisational policies, including professional development programmes, specialised advice and the evaluation of digital accessibility as a criterion for educational quality. Pedagogical leadership and institutional coordination are essential to consolidate a culture of sustainable digital inclusion at all levels of the education system.

Finally, perceived inclusive digital competence should be conceived as a structural indicator of equity, incorporated into innovation, teacher evaluation and institutional accreditation policies. Only through a systemic and shared vision will it be possible to reduce the gaps detected and promote a truly inclusive digital education model.

Limitations and future lines of research

Although the instruments used were equivalent and demonstrated high levels of reliability and validity, the difference in the number of items between the two versions may have introduced slight variations in the comparison between groups. Nevertheless, the confirmed scalar invariance and the consistency of the results obtained following standardisation support the robustness and comparability of the conclusions.

Furthermore, the non-probabilistic and purposive nature of the sampling should be considered a limitation, as should the imbalance in the size of the samples analysed, with the sample of university lecturers being considerably larger than that of primary school teachers. In addition, participation was voluntary; it is therefore possible that teachers with a greater interest in educational technology and inclusion were more inclined to complete the questionnaire. These factors limit the representativeness of the sample and suggest that the results should be interpreted with caution when it comes to generalising.

Furthermore, the study relies exclusively on self-report measures and assesses teachers' perceived inclusive digital competence. Although this approach is useful for understanding professional beliefs and perceptions, it does not guarantee that these perceptions correspond to the competences actually demonstrated in teaching practice. Responses may be influenced by factors such as social desirability, over- or under-confidence, or various response biases. Consequently, the results should be interpreted as indicators of perceived competence rather than as direct evidence of actual professional performance.

Similarly, inclusive digital competence is a complex, multidimensional and dynamic construct. Although the factor analyses carried out support the internal structure of the instruments, the questionnaires used represent a specific operationalisation of the construct and may not capture all the dimensions involved in inclusive teaching practice. In particular, aspects relating to the effective implementation of Universal Design for Learning, the accessibility of digital materials or the actual use of assistive technologies may require complementary assessment procedures.

Furthermore, the study does not incorporate direct observations of teaching practice or objective analyses of the digital resources used by teachers. It is therefore not possible to determine to what extent the perceived competences translate into specific inclusive behaviours, nor to assess their actual impact on pupils' learning opportunities. Furthermore, the cross-sectional nature of the research design should be considered a limitation. Although this approach allows associations and differences between groups to be identified at a given point in time, it does not allow causal relationships to be established or the direction of the observed effects to be determined. Consequently, the results should be interpreted as evidence of an association between variables and not as causal explanatory relationships.

Also, although statistically significant differences were identified between the groups analysed, the effect sizes observed were generally small or moderate. Similarly, the explanatory power of the predictive models was limited (R^2 = .112 in primary education and R^2 = .127 in higher education), indicating that a significant proportion of the variability in perceived inclusive digital competence remains unexplained. This finding suggests the existence of additional institutional, organisational, technological and cultural factors that were not incorporated into this study and which could play a significant role in shaping these competences.

Looking ahead, it is proposed to extend the research to other educational levels, particularly secondary education and vocational training, and to incorporate mixed-methods designs that combine teachers' self-perceptions with observations of teaching practice, analysis of digital materials and objective measures of performance. Furthermore, it would be appropriate to develop longitudinal, experimental or quasi-experimental studies to analyse the development of inclusive digital competence and assess the impact of specific training programmes on teaching practice.

Finally, future research should explore the relationship between teachers' inclusive digital competence and pupils' learning outcomes, participation and well-being – particularly those of pupils with disabilities – in order to gain a better understanding of the actual impact of these competences on educational equity.

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